

R1-Zero and Pure RL Reasoning

Deep Dive into DeepSeek — Chapter 8

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Foundation Books Project

The one rule of this chapter

A rule cannot optimize properties that it never measures.

- DeepSeek-R1-Zero applies large-scale reinforcement learning directly to a base model, DeepSeek-V3-Base, with **no** supervised cold-start stage first.
- A base policy samples completions; an extractor isolates a gradable answer; a rule-based verifier turns that answer into reward.
- Within-prompt comparisons yield group-relative advantages, and a guarded update moves the policy toward what the group rewarded.

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Goal: follow one small congruence example through reward, group comparison, and update pressure, then read the reported R1-Zero behaviors — and their limits — against that mechanism.

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**From a base policy to a
rule-verifiable reward**

The running example: one prompt, four rollouts

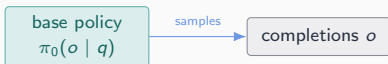
Find $n \in \{0, \dots, 34\}$ such that $n \bmod 5 = 2$ and $n \bmod 7 = 3$. Unique answer: 17.

base policy
 $\pi_0(o \mid q)$

- A base policy $\pi_0(o \mid q)$ samples completions o for prompt q . An **extractor** E maps a completion to a gradable answer or to $\emptyset$. A **verifier** $V(q, E(o))$ returns $\{0, 1\}$.
- The verifier scores only the extracted object — not the full completion, and not an intuition about whether the prose looks plausible.

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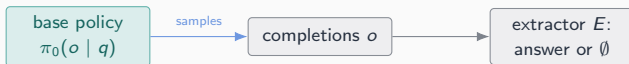
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This asymmetry — cheap to check the answer, hard to check the reasoning — is the seed of both this chapter's success story and its failure modes.

Accuracy, format, and total reward stay separate fields

Candidate	Extracted answer	$r_{\text{acc}}, r_{\text{fmt}}$	r
o_1	17, tagged	1, 1	1.1

$$r(q, o) = r_{\text{acc}}(q, o) + \lambda r_{\text{fmt}}(o), \text{ toy weight } \lambda = 0.1.$$

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o_1	17, tagged	1, 1	1.1
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o_4	17, untagged	1, 0	1.0

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Correctness and formatting are different objects. Neither implies the other — o_2 is formatted but wrong, o_4 is right but unformatted.

GRPO: group-relative comparison

The group supplies its own baseline

Definition (Group-relative advantage)

For a prompt q , sample G completions from $\pi_{\theta_{\text{old}}}$. With rewards $r_1, \dots, r_G$ and $\sigma_r \neq 0$,

$$A_i = \frac{r_i - \mu_r}{\sigma_r}, \quad \mu_r = \frac{1}{G} \sum_j r_j.$$

When $\sigma_r = 0$, set $A_i = 0$ for every i .

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No trained critic supplies this baseline — the sampled group does. DeepSeekMath motivates GRPO by removing the value model; the group still costs G rollouts and G reward evaluations per prompt.

The PPO-to-GRPO bridge

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Update direction	Learned critic, often with GAE	Within-prompt group-relative reward comparison

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GRPO trades a trained critic for rollout cost. It keeps the clipped-ratio guard exactly where PPO put it.

The objective, KL placement, and token tensors

One objective, three separable jobs

$$\ell_i^{\text{seq}}(\theta) = \min\left(\rho_i^{\text{seq}} A_i, \text{clip}(\rho_i^{\text{seq}}, 1-\epsilon, 1+\epsilon) A_i\right) - \beta D_{\text{KL},i}^{\text{seq}}$$

- $\rho_i^{\text{seq}} = \pi_{\theta}(o_i \mid q) / \pi_{\theta_{\text{old}}}(o_i \mid q)$ measures movement; A_i supplies direction and strength.
- Task rewards form A_i **first**. The KL term to π_{ref} is subtracted **after**, directly in the loss — it never changes the group's reward mean, spread, or advantage signs.

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A reproducible record must say *actor-loss KL* or *in-reward KL*: public trainers use either, and only one matches this paper-level placement. [REPORTED: DeepSeek-R1 paper]

Token tensors: same terms, different weight

Rewards/advantages: shape $[G]$. Token ids, ratios, mask: shape $[G, T]$.

- Response-token loss: $\ell_i^{\text{tok}} = \frac{1}{M_i} \sum_t m_{i,t}[\dots]$ — variable per-response weight $1/M_i$.
- Dr. GRPO's fixed-divisor variant instead uses $1/T_r$, a shared budget — and drops the σ_r standardization in A_i .
- Equal-response-weight and equal-valid-token-weight reductions of the *same* masked terms gave $J_{\text{response}} \approx 0.024$ versus $J_{\text{valid token}} \approx -0.107$ in the chapter's worked trace.

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A reported GRPO loss number is uninterpretable until its reduction is named. [DERIVED: worked token trace, this chapter]

What emerged, and what broke

Reward pressure correlates with reasoning-shaped behavior

- Reported: accuracy improves through training; average response length **increases** on the training set.
- Reported behaviors: self-verification, reflection, longer chains of thought, exploring alternative approaches — including a notable “aha moment” of mid-solution revisiting.
- AIME 2024 pass@1: 15.6% → 71.0%; majority voting reaches 86.7%. [REPORTED: DeepSeek-R1 paper]

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These are operational, observational claims about sampled outputs under the reported setup — not evidence about model intent, and not proof that every long trace is useful.

The reward's blind spots survive the pressure

Failure mode	Why the reward allows it	R1 bridge
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R1-Zero has reported drawbacks: poor readability and language mixing. The repository README separately warns about repetition and incoherent output in usage guidance.

[REPORTED: DeepSeek-R1 paper and README]

**Fit, evidence, and what carries
forward**

When pure rule RL is the right tool

- Prefer it when outcomes are reliably verifiable, the reward has been tested for shortcuts, and the base policy already produces enough successes for group contrast.
- Readability, language consistency, and robustness to hidden variants stay outside the reward — they need cold-start data, filters, preference signals, or a stronger teacher.
- A rule-verifiable final artifact (an integer, a passing test suite) does not certify everything about the surrounding behavior.

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DeepSeek-R1 adds exactly what pure rule RL cannot supply on its own: cold-start data, rejection-sampled SFT, and staged RL — Chapter 9.

The chapter's own arithmetic, reproduced locally

- `code/chapter_r1_zero_pure_r1_reasoning/toy_reward_and_grpo.py` prints `mean=0.55`, `population_std=0.502494`, and `pass_at_one=0.6875`.
- Those are exactly this chapter's worked mean, standard deviation, and the 16-sample congruence `pass@1` check — recomputed, not asserted.

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This is the book's own companion harness, not a DeepSeek repository excerpt: it lets every claimed number in the worked example be rerun rather than trusted. [DERIVED: this chapter's companion harness]

Concept-to-code map for the chapter

- R1-Zero training regime (no cold start) → [DeepSeek-R1/README.md](#)

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This chapter's public sources are papers, a model card, and one docs page — not an executable DeepSeek trainer. The DeepSeek-R1 chapter (9) is where those same ideas meet real veRL configuration objects.

What carries forward

Sample a group. Score it with a rule. Compare within the group.
Move the policy, guarded — and audit what the rule never measured.

- The mechanism is now fully computable: reward, advantage, ratio, clip, KL placement, token mask, reduction.
- The limit is equally load-bearing: rule rewards see only what extraction and verification expose.

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Next: Chapter 9 keeps this GRPO mechanism and adds cold-start data, rejection-sampled SFT, and staged RL to reach a usable, readable model family — DeepSeek-R1.